



DPP-2021-West Phase-2 System Impact Study Report - FINAL

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MISO

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Executive Summary

This report presents the results of a System Impact Study performed to evaluate the interconnection of the generators in the Definitive Planning Phase (DPP) 2021 West Phase 2. The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models, and results. The ad hoc study group consisted of representatives from the interconnection customers and the utility companies outlined in Table 1 and Table 2.

Table 1: MISO Transmission Owners

MISO Transmission Owner
Dairyland Power Cooperative
Great River Energy
ITC Midwest
MidAmerican Energy Company
Minnesota Power
Missouri River Energy Services
Montana Dakota Utilities
Muscatine Power & Water
Northern States Power (Xcel Energy)
Otter Tail Power Company
Southern Minnesota Municipal Power Agency

Table 2: Affected Systems

Affected System
Associated Electric Cooperative Inc.
Manitoba Hydro
Minnkota Power Cooperative Inc.
Saskatchewan Power
Southwest Power Pool
PJM ¹

53 interconnection requests were studied with a combined nameplate rating of 7,797.4 MW for Energy Resource (ER) and 7,376.43 MW of Network Resource (NR) service. Non-CEII information such as fuel type, project location, and Point of Interconnection may be obtained through the [MISO Public Queue](#). Full project details and modeling assumptions used in this analysis are provided in Appendix J.

¹ PJM preliminary report should be considered as non-binding, information only for the purposes of demonstrating the result under the proposed new study methodology in the JOA revision which is pending FERC approval. PJM will issue the formal AFS report after the JOA is approved and move forward from there.



Table 3: Interconnection Requests Under Study

Project Number	POI NAME	Service Type	Fuel Type	Fuel Breakout	ERIS	NRIS	Total Network Upgrade Cost for Milestone Calculation
J1846	Panther 230kV Substation	NRIS	Solar		200	200	\$ 27,521,456.94
J1847	Murphy Creek 161kV substation	NRIS	Solar		200	200	\$ 17,826,820.09
J1867	Quinn 345kV	NRIS	Wind		200	200	\$ 24,283,096.30
J1888	SQUARE BUTTE EAST 230kV	NRIS	Wind		175	175	\$ 49,937,445.86
J1889	Mt Vernon 161kV Substation	NRIS	Solar		60	60	\$ 1,489,622.53
J1893	Mandan 230kV	NRIS	Wind		500	500	\$ 198,308,321.36
J1898	BYRON 161kV	NRIS	Battery Storage		200	200	\$ 13,989,147.70
J1903	Osceola 161 kV	NRIS Only	Wind		0	140	\$ 2,148,462.76
J1927	Grandview Substation 69kV	NRIS	Solar		24	24	\$ 840,603.63
J1952	Emery-Floyd 161kV	NRIS	Wind		180	54	\$ 88,205,389.58
J1961	Buffalo 115kV	NRIS	Wind		190	57	\$ 78,782,989.21
J1978	Lemars-Little Sioux 161kV	NRIS	Wind		170	52.5	\$ 36,282,497.07
J1979	Hazleton -Mitchell County 345kV	NRIS	Wind		200	60	\$ 134,798,113.90
J1989	Twin Brooks-Ellendale 345kV	NRIS	Wind		250	75	\$ 117,547,900.43
J1990	Ellendale 345kV	NRIS	Wind		215	75	\$ 81,577,497.63
J1998	Nuthatch Switching Station 161kV	NRIS	Wind		66	66	\$ 27,684,910.97
J2002	345kV Sub T Substation	NRIS	Solar		200	200	\$ 11,796,885.00
J2021	Tremval 69kV	NRIS	Solar		48.43	48.43	\$ 5,092,435.08
J2022	Pleasant Valley 345kV Substation	NRIS	Solar		200	200	\$ 36,832,636.13
J2027	Chub Lake-Hampton Corners 345kV	NRIS	Solar		150	150	\$ 13,217,130.07
J2030	ISANTTI-ATHNSSW 69kV	NRIS	Solar		56	56	\$ 21,769,784.48
J2031	161kV Toledo Substation	NRIS	Wind		100	100	\$ 10,911,110.38
J2040	Lakefield-Obrien Co 345kV	NRIS	Wind		150	150	\$ 36,776,096.18
J2053 ²	Lyon County-Cedar Mountain 345kV	NRIS	Wind		151.47	122	\$ 37,626,710.06
J2054	Freeborn 161kV	NRIS	Wind		67	67	\$ 78,793,311.30

² The ICIF report is currently unavailable



J2065	Buffalo Ridge-Pipestone 115kV	NRIS	Solar		200	200	\$	18,489,219.29
J2066	Denmark 161kV	NRIS	Battery Storage		100	100	\$	15,584,780.20
J2068	Herbert 69kV	NRIS	Solar		75	75	\$	4,324,571.27
J2070	Three Lakes-Pine Lake 115kV	NRIS	Solar		100	100	\$	15,613,330.05
J2078	Fernald Substation 161kV	NRIS	Solar		200	200	\$	4,223,273.11
J2079	Lime Creek 161kV	NRIS Only	Wind		0	150	\$	2,951,145.14
J2080 ³	Square Butte East 230kV	NRIS	Wind		200	200	\$	57,553,732.45
J2084	LINTON-ESTBMRK 115kV	NRIS	Solar		45	45	\$	32,813,897.55
J2085	LINTON 115kV	NRIS	Solar		40	40	\$	30,216,827.36
J2086	Buffalo 345kV	NRIS	Solar		500	500	\$	62,657,331.15
J2087	Franklin 115kV	NRIS	Solar		125	125	\$	11,368,642.72
J2097 ²	Tri County 230kV Substation	NRIS	Wind		100	100	\$	16,483,931.34
J2101	Mandan 230kV	NRIS	Battery Storage		100	100	\$	24,151,514.74
J2102	Alma 161kV	NRIS	Battery Storage		150	150	\$	12,951,159.92
J2104	Lyon County-Cedar Mountain 345kV	NRIS	Wind		230	230	\$	41,275,418.51
J2106	Story County 161kV	NRIS Only	Wind		0	150	\$	1,760,083.65
J2116	Colvill 69kV	NRIS	Diesel		18	18	\$	1,728,209.02
J2120	Fairfax 161kV	NRIS	Battery Storage		100	100	\$	24,585,355.92
J2122	Nuthatch Switching Substation 161kV	NRIS	Wind		33	33	\$	11,031,719.71
J2157 ²	Tri County 230kV Substation	NRIS	Wind		100	100	\$	12,384,565.37
J2199	COON CREEK 345kV	NRIS	Battery Storage		200	200	\$	25,184,274.71
J2219	Rochester-Wabaco 161kV	NRIS	Solar		180	180	\$	12,642,910.97
J2227	OSSEO-STRUM 69kV	NRIS	Solar		89	89	\$	28,858,010.68
J2250	Ellendale sub 345kV	NRIS	Wind		200	200	\$	83,583,730.11
J2268	ARNOLD 345kV	NRIS	Solar		370	370	\$	17,305,622.44
J2272	East Calamus 161kV Substation	NRIS	Hybrid	Solar/Battery	250	250	\$	3,435,386.99
J2276	Alexandria Tap 115kV	NRIS	Solar		39.5	39.5	\$	4,305,325.57
J2309	Chanarambie-Ellsborough 115kV	NRIS	Solar		100	100	\$	12,737,644.51

³ Interconnection Service for this project depends on the continued existence of the Square Butte HVDC TSR.



The full list of network upgrades required are detailed in Appendix B. The cost responsibility for these upgrades has been allocated in accordance with MISO BPM-015. Please refer to Appendix A for the breakdown cost for each Interconnection Request and the associated milestone payments required to advance to the next phase.

Model Development

The origin of the DPP2021 models is the MTEP21 Target A models with the Bench Cases including all prior-queued projects, while the Study Cases contain all the interconnection requests in DPP2021, in addition to all the facilities in the Bench Cases.

Model Development and Study Assumptions

Base Case Models

The origin of the DPP 2021 Phase II models is the MTEP21 models with the Bench Cases including all pre-queued projects and associated network upgrades, while the Study Cases contain all the interconnection requests in DPP 2021 Phase II, in addition to all the facilities in the Bench Cases. Below is the list of different scenarios.

- Bench Cases
 - DPP21-2026SH-Phase II-Bench
 - DPP21-2026SUM- Phase II-Bench
- Study Cases
 - DPP21-2026SH- Phase II-Discharging
 - DPP21-2026SUM- Phase II-Discharging
 - DPP21-2026SH- Phase II-Charging

Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the <DPP Study Cycle Phase I/II/III> generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 34.9.3 and TARA version 2301_2



Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent conditions (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 "Reactive Power Requirements for Non-Synchronous Generation", which was issued June 16, 2016, states that "Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation." As such, this Final Rule applies to all wind and solar projects included in this study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter-based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system). Results of the FERC Order 827 Compliance Review are detailed in Appendix K.



Steady State Thermal Analyses

The steady state ERIS and NRIS (deliverability) analyses are conducted to identify what network upgrades, if any, are required to reliably interconnect current study interconnection requests to the MISO transmission system. Section 6 of the MISO BPM-015 details the processes and methodologies used for the DPP steady state thermal analysis.

ERIS Analysis

The following models were developed for the ERIS analysis:

- Summer Shoulder Bench
- Summer Shoulder Discharging Study
- Summer Shoulder Charging Study
- Summer Peak Bench
- Summer Peak Discharging

The origin of the DPP21 Study Cycle Phase II models is the MTEP21 models with the Bench Cases including all pre-queued projects and associated Network Upgrades, while the Study Cases contain all the interconnection requests in DPP21 Study Cycle Phase II, in addition to all the facilities in the Bench Cases.

MISO utilized PSS®E 34.9.3 and PowerGEM TARA v2301_2 for the ER analysis. AC Contingency calculations were performed on the bench and study models and identified network constraints in accordance with the MISO BPM-015. The full list of constraints identified in the ERIS analysis is detailed in Appendix C.

Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

Contingencies

The following contingencies were considered in the steady state analysis:

1. NERC Category P0 (system intact – no contingencies)
2. NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
3. NERC Category P2-P7 contingencies
4. For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

Solutions to constraints observed in the system impact study were developed in partnership with the host and affected transmission owners. Network upgrades were allocated cost in accordance with the MISO BPM-015. The list of network upgrades required for interconnection service are detailed in Appendix B. Cost allocation of those upgrades along with the required milestone payments to advance to the next phase are outlined in Appendix A.

Network upgrade cost estimates were developed on a planning level and are later refined in the Phase II studies. Details pertaining to upgrades, costs, and the execution plan for interconnection



of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator. Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.

NRIS (Deliverability) Analysis

Generator interconnection projects must pass the Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource project or to proceed with the system upgrades that will make the generator fully deliverable. The Generator Deliverability Study ensures that Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak conditions without getting bottled up. The wind and solar generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

The MISO Generator Deliverability Study whitepaper describing the algorithm can be found in Appendix C of BPM 015.

The model developed for the NRIS analysis was derived from the Summer Peak Discharging case from the ERIS model; however, the study units are not dispatched as this occurs through the TARA deliverability analysis. The full list of constraints identified in the NRIS analysis is detailed in Appendix D.

Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all its output is not deliverable. The restricted MW is calculated as follows:

$$(MW\ restricted) = \frac{worst\ loading - MW\ rating}{(generator\ sensitivity\ factor)}$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

Shared Network Upgrades Analysis

The Shared Network Upgrade (SNU) Analysis, which tests Network Upgrades driven by higher queued interconnection projects, was performed for this System Impact Study. Please refer to Appendix C for maximum MW impacts and SNU cost allocations.

Stability Analysis

Stability analysis is performed on the study units to determine their dynamics performance. Each region performs stability analysis using the dispatch from the ERIS case. Supplemental analysis may also be performed to capture relevant Local Planning Criteria (LPC) for new generator interconnections.



Model Development

Stability models are developed using the ERIS five year out summer peak model. LPC stability models are developed based on the stability study models and adjusted to comply with the relevant LPC by fully dispatching nearby local generation.

- Bench Case
 - DPP21-2026SH90-Phase II-Bench.pfb
 - DPP21-2026SUM-Phase II-Bench.pfb
- Study Case
 - DPP21-2026SH90-Phase II-Study_Discharging.pfb
 - DPP21-2026SH90-Phase II-Study_Charging.pfb
 - DPP21-2026SUM-Phase II-Study_Discharging.pfb

Study Methodology

The purpose of the study is to identify potential angular instabilities, voltage dip violations, and damping violations, if any, due to the interconnection of the projects in the study cycle under disturbance conditions, and the impact of all study projects on the system stability performance.

The fault scenarios simulated in this study cover faults simulated as part of the MTEP 21 analysis as well as selected three-phase (3PH) faults with normal clearing and single line to ground (SLG) faults with delayed clearing. Dynamic simulations of fault scenarios were performed using the DSA Tools TSAT program (version 24).

Fault scenarios were first simulated using the study case and the results were reviewed. For scenarios that exhibited instability, the bench case was simulated such that the stability performance with and without the proposed interconnection projects could be compared. Any new stability problems attributed to the proposed interconnection projects are flagged and reported.

For each fault, rotor angles, speed deviation, and electrical power outputs of the study generators and the generators in the proximity were monitored. Voltages at selected buses, including all POI buses of the study projects and all future buses, were also monitored.

In addition, all renewable study projects are subject to the voltage ride-through, and frequency ride-through criteria specified in NERC PRC-024-2 ("Generator Frequency and Voltage Protective Relay Settings"⁴) to check if the projects remain connected during frequency and voltage excursions. Specifically, PRC-024-2 mandates that protective relaying should be set in such a way that:

- Voltage Ride-Through: a generator shall withstand zero voltage at the POI (typically the primary side of the transformer station) for up to 0.15 seconds (9 cycles) and the ensuing voltage recovery period for three phase faults.
- Frequency Ride-Through: a generator shall maintain continuous operation between 59.5 and 60.5 Hz.

⁴ <http://www.nerc.com/pa/Stand/Reliability%20Standards/PRC-024-2.pdf>



Local Planning Criteria

Otter Tail Power Company

Otter Tail Power (OTP)'s Local Planning Criteria (LPC) analysis is being performed due to interconnections J1961, J1989, and J2086 requesting interconnection into OTP's transmission system. The OTP LPC analysis consisted of steady-state contingency analysis for Summer Shoulder (SH), Summer Peak (SPK), and Light Load No Wind (LLNW) conditions.

1. OTP LPC SPK
2. OTP LPC SH
3. OTP LPC LLNW

Any constraint or violation of system performance criteria that occurs in the LPC Study Cases and does not occur in the LPC Benchmark Cases should be identified, allocated and mitigated by the 2021 DPP generators. OTP LPC analysis results are detailed in Appendix F

Great River Energy

The Great River Energy (GRE) Local Planning Criteria (LPC) study is performed to adhere to GRE Local Planning Criteria. This LPC analysis is required for J1846, J2022, and J2030 projects in addition to MISO's standard DPP analysis. The GRE LPC analysis consisted of steady-state contingency analysis for Summer Peak (SPK) and Summer Shoulder (SH) conditions. Individual models were developed for each project J1846, J2022, and J2030.

1. GRE LPC SPK
2. GRE LPC SH

Any constraint or violation of system performance criteria that occurs in the LPC Study Cases and does not occur in the LPC Benchmark Cases should be identified, allocated and mitigated by the 2021 DPP generators. GRE LPC analysis results are detailed in Appendix F.

Dairyland Power Cooperative

The Dairyland Power Cooperative (DPC) Local Planning Criteria (LPC) study is performed to adhere to DPC Local Planning Criteria. This LPC analysis is required for J2102, J2219, and J2227 projects in addition to MISO's standard DPP analysis. The DPC LPC analysis consisted of steady-state contingency analysis for Summer Peak (SPK) and Summer Shoulder (SH) conditions. Individual models were developed for each project J2102, J2219, and J2227. A SPK case was not required for J2227.

- 1) DPC LPC SPK
- 2) DPC LPC SH

Any constraint or violation of system performance criteria that occurs in the LPC Study Cases and does not occur in the LPC Benchmark Cases should be identified, allocated and mitigated by the 2021 DPP generators. DPC LPC analysis results are detailed in Appendix F.

Montana-Dakota Utilities Company

Montana-Dakota Utilities Company (MDU) determined that J1990 has more than 20% DF on the outlets of J436/J437/J488, J2084 has more than 20% DF on the outlets of J1040, and J2085 has more than 20% DF on the outlets of J1040 and G359 Merricourt Wind. In addition to MISO's standard DPP analysis, MDU determined that J1893, J1990, J2084, J2085, J2101, and J2250 are required for MDU LPC study per Section 3.2 of the MDU Local Planning Criteria. The MDU LPC analysis consisted of steady-state contingency analysis for Summer Peak (SPK) and Summer Shoulder (SH) conditions.

- 1) MDU LPC SPK
- 2) MDU LPC SH

Any constraint or violation of system performance criteria that occurs in the LPC Study Cases and



does not occur in the LPC Benchmark Cases should be identified, allocated and mitigated by the 2021 DPP generators. MDU LPC analysis results are detailed in Appendix F.

Short Circuit Analysis

Short circuit analysis was performed utilizing CAPE software. Single line to ground faults and three phase faults were evaluated for pre- and post-project cases (comparable to Bench and Study models). The details pertaining to the short circuit analysis are presented in Appendix G.

Affected System Impact Study

The details pertaining to the Affected Systems studies are in Appendix H.



Appendices

Appendix A - Cost Allocation (CEII)

Appendix B - Network Upgrade Summary (CEII)

Appendix C - ERIS Analysis (CEII)

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Appendix L - Voltage Stability Assessment (CEII)